DESIGN SPECIFICATION / EXEMPLAR 01

Relational Process Framing

A Forkable Descriptive Generalism with a Causal-Resolution Exemplar

Subdocument to the broader process-description and simulation toolset

CORE PROPOSITION

Descriptions may remain freely authored. Once rules bind relations within a declared frame, those rules govern what that frame may coherently describe as happening next.

Document role	Focused project subdocument; one generalism and one worked exemplar.
Current exemplar	Finite-speed causal resolution across space and time.
Design posture	Permissive toward fiction and speculation; explicit about axioms, scope, provenance, and consequence.
Version	Draft 0.1 - August 2026

This document defines a usable framing method. It does not redefine the whole project around this single contemplation.

Contents

Navigation note

This focused subdocument uses numbered headings and a compact section guide. PDF page thumbnails and search provide additional navigation.

Executive Summary

This subdocument isolates one general method supported by the broader toolset and develops one demanding exemplar. The method treats a description as a configurable arrangement of identities, states, relations, transformations, events, frames, and observations. The tool may host ordinary stories, process diagrams, scientific models, mathematical structures, speculative worlds, or deliberately absurd accounts. It does not require all descriptions to resolve into one ontology or one executable universe.

Freedom of description remains the starting condition. Constraint appears when a user assigns rules to relations inside a declared frame. A rule can govern what may interact, what must wait, what persists, what counts as evidence, which transformations remain possible, and which outcomes become contradictory under that frame. Because frames and their axiom sets can branch, the same scenario may be explored under different assumptions without silently treating one branch as the only meaningful account.

The worked exemplar concerns causal resolution. It shows how a story or simulation can include spatially and temporally governed relations, including finite propagation speed, while also containing atemporal, aspatial, symbolic, interpretive, or explicitly instantaneous relations. These modes do not automatically produce a paradox. They become contradictory only when the same scoped claim is required to obey incompatible rules at once. Explicit cross-frame bindings determine when one mode can affect another.

This design lets a user compare descriptions across varied axioms. A fictional branch may hold greater artistic, emotional, or conceptual value than a physically realizable branch. The system does not flatten those values into one ranking. At the same time, it can identify the exact step at which a claim leaves a declared scientific or mathematical basis, enters an added axiom, invokes a narrative override, or remains unresolved. The result is neither unrestricted hand-waving nor compulsory reduction to one physical model. It is an inspectable environment for process framing.

> **GENERAL RULE**
>
> *The environment permits descriptions freely; declared relational rules determine how those descriptions may develop, interact, resolve, or remain unresolved within their stated scope.*

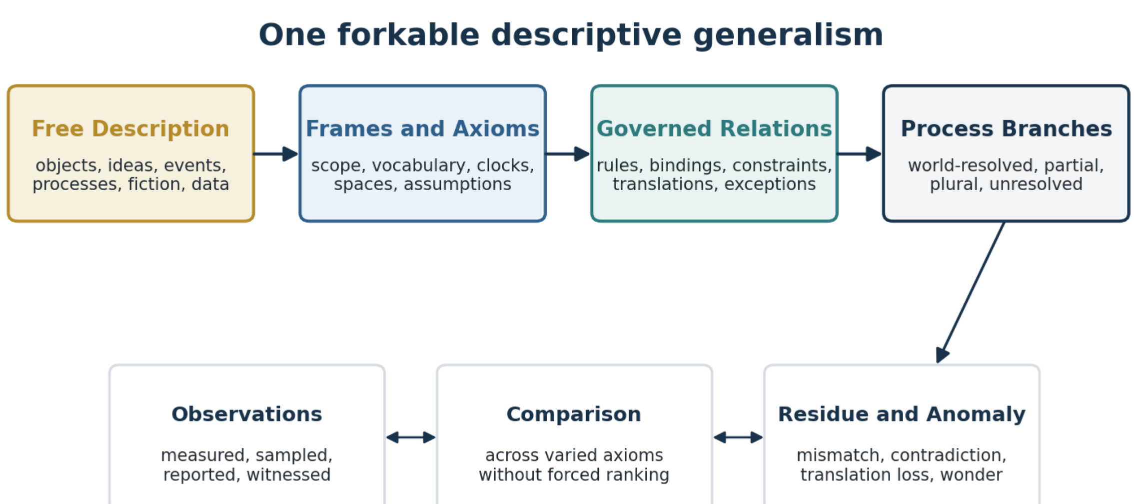

Figure 1. The current generalism: free description becomes operational through declared frames, axioms, and governed relations. The generalism itself may later be forked.

1. Document Position and Scope

1.1 A subdocument, not a project reorientation

The broader project provides a toolset for authoring, organizing, visualizing, executing, and revising descriptions of processes. This document does not redefine that whole effort as a physics engine, a consciousness theory, a model-comparison project, or a philosophical investigation into persistence. Each of those may become a use case, a native vocabulary, or a future exemplar.

This subdocument instead records one useful generalism: a process can be framed through identities and relations, and rules attached to those relations can govern permitted continuations. The causal-resolution example demonstrates that the same generalism can accommodate unusually strict spatial and temporal constraints without losing the capacity for fiction, symbolic relations, or unresolved alternatives.

1.2 What this document commits to

- A description may be physical, fictional, mathematical, cognitive, biological, symbolic, narrative, incomplete, contradictory by design, or mixed.
- A frame declares the local vocabulary, assumptions, clocks, spaces, relation types, and standards of consequence that apply to a description.
- A branch inherits a frame and may override or replace axioms without erasing its parent.
- A rule gains operative force when it binds a relation or relation class within a scope.
- Cross-frame interaction requires an explicit bridge, translation, observation, or override.
- Comparison records agreement, mismatch, residue, anomaly, and value without imposing one universal ranking.
- The current physical exemplar remains an example of the method, not its final or exclusive form.

1.3 What remains outside this subdocument

This document does not specify the complete editor, storage layer, rendering engine, collaboration model, asset pipeline, agent architecture, or final user interface of the larger project. It defines the conceptual and operational requirements needed for this one framing method to integrate cleanly with those systems.

> **Design boundary**
> The document is complete when it explains how free descriptions acquire scoped consequences, how distinct relation classes coexist, and how the causal exemplar operates. It need not turn every future use case into a full theory.

2. The Forkable Descriptive Generalism

2.1 Descriptions as process framings

A description becomes a process framing when it specifies, even minimally, what can be distinguished, how those distinctions relate, what can change, and how a later state or interpretation may follow from an earlier one. The framing may describe a world, but it need not. It may instead describe a proof, a design, a memory, a symbolic sequence, a musical development, an argument, or a narrative possibility.

The minimal pattern is:

```
Distinctions or identities
        +
Relations among them
        +
Possible transformations
        +
A frame that states where the terms apply
        =
A process description that may be inspected, narrated, or executed
```

This structure remains intentionally weaker than a universal ontology. It does not decide whether objects are fundamental, whether relations are fundamental, whether time is continuous, whether observers are internal to the system, or whether every description refers to one reality. Those commitments belong to frames and branches.

2.2 The generalism is itself forkable

The current generalism uses identities, states, relations, transformations, events, frames, and observations because these primitives support a broad range of work. The system should nevertheless preserve the possibility of forking the generalism itself. A later branch might reject persistent identity, replace state transitions with constraint satisfaction, treat events as derived rather than primitive, or organize descriptions through categories, sheaves, fields, or pure relations.

Forkability at this level matters because the tool should not claim neutrality while secretly making its present data model irreversible. The current primitives should function as a strong working basis, not as an unchallengeable metaphysics.

2.3 Process, narrative, and world

A narrative is one important form of process description. It usually combines distinguishable participants, ordered occurrences, changing relations, remembered conditions, and consequences. A world model is a more constrained form that supplies enough rules to calculate or validate some of those changes. The two overlap, but neither reduces cleanly to the other.

Description mode	Primary question	Typical commitment
Story or mini-narrative	What happens, to whom, and with what significance?	Authored order and meaning; may omit mechanism.
Process description	How do distinctions, relations, and transformations develop?	Traceable change; may remain partly symbolic.
World model	What states and events follow under declared rules?	Operational consequence and conflict resolution.
Formal description	What follows from definitions or axioms?	Derivation; may be atemporal and aspatial.
Interpretive layer	What does an event mean to an observer or author?	Significance without automatic physical force.

2.4 Native structural questions

The project may expose a recurring set of questions as native relation and inspection categories. These questions originate naturally in biological, cognitive, cybernetic, and systems-oriented work, but they remain useful beyond those domains:

Operation	Emergent concern	Question
Distinction	**Identity**	What is it?
Transformation	**Change**	What is it doing?
Organization	**Function**	How does it work?

Operation	Emergent concern	Question
Continuity	**Persistence**	How long can it hold together?
Invariance	**Regeneration**	How can its pattern reappear?

3. Free Description and Accountable Constraint

3.1 Fiction is not a defect state

The environment should permit a user to author impossible geometry, instantaneous knowledge, retrocausal prophecy, conscious architecture, symbolic causation, a talking star, or a universe whose laws change during a scene. A story can carry emotional, aesthetic, conceptual, or personal value that exceeds the value of a physically realizable account for the person creating it. The system should not respond by assigning fiction a lower global status.

What the system can provide is accountability. It can show which claims are authored, which are derived, which are measured, which rely on an added axiom, which violate a selected physical pack, and which remain meaningful only inside a narrative or interpretive frame. This preserves expressive freedom while making the path of consequence visible.

PERMISSIVE AUTHORING RULE

The system does not forbid invention. It distinguishes the source, scope, and consequence of each invention.

3.2 Rules create local necessity

A rule should not merely annotate a description. Once activated, it constrains the relations within its declared domain. A finite-propagation rule can prevent a detector response before signal arrival. A trust relation can make one character act on another character's report. A logical rule can make a conclusion derivable. A narrative decree can establish that an event occurs even when no mechanism has been modeled.

These rules do not all operate in the same manner. The engine must therefore retain their type and scope. A rule may govern physical transition, narrative order, interpretation, formal entailment, interface behavior, or cross-frame translation.

Rule class	Binds	Produces	Does not automatically produce
Physical	states, channels, distances	eligible state changes	meaning or narrative priority
Narrative	event order, roles, significance	story continuation	a physical mechanism
Interpretive	observer, evidence, belief	updated meaning or policy	retroactive world-state change
Formal	symbols, propositions, axioms	derivation or contradiction	spatial movement
Cross-frame	source and target descriptions	translation or intervention	identity of the two frames

3.3 The accountability boundary

For users who want scientific or mathematical accountability, every consequential chain should expose the point at which its status changes. A result may begin with measured values, proceed through an accepted equation, invoke a numerical approximation, cross a translation mapping, and finally depend on an invented axiom. The tool should mark those transitions rather than presenting the entire chain as either equally established or equally fictional.

```
Measured input
  -> declared model
  -> calculated propagation
  -> sampled state
  -> interpolation
  -> cross-frame translation
  -> authored exception
  -> final narrative event
```

This chain can remain valuable at every stage. The status markers answer different questions: where does empirical accountability end, where does mathematical derivation continue, where does approximation enter, and where does creative authorship take control?

3.4 Value is multidimensional

The comparator should avoid one universal score. A branch can instead carry a profile across several dimensions:

- Internal coherence under its own axioms.
- Mathematical derivability or formal precision.
- Compatibility with a selected physical model.
- Empirical traceability to observations.
- Narrative power, emotional value, or symbolic importance.
- Explanatory economy or conceptual usefulness.
- User-declared priority for the present task.

> **No automatic demotion**
> A fictional branch may score low on physical compatibility and high on narrative value. A strict physical branch may reverse that profile. The system records the difference; it does not decide which value the user ought to prefer.

4. Frames, Axioms, and Branches

4.1 Frames define the local terms of consequence

A frame states where a vocabulary and rule set apply. It may define the available identities, the kinds of relation that can connect them, whether space or time applies, which clocks or coordinate systems are used, what counts as an event, what qualifies as evidence, and how conflicts are handled.

Frames prevent the environment from treating every statement as a claim in one undifferentiated universe. A mathematical implication can remain valid without traveling through space. A memory can influence an agent's choice without becoming a force vector. A narrator can reveal information instantly to the audience while the characters remain physically isolated.

4.2 Axioms are explicit commitments

An axiom set may include established physical laws, modeling assumptions, fictional premises, computational simplifications, narrative conventions, or interpretive policies. The system should preserve the difference between inherited axioms and local overrides.

```
Frame: Laboratory Physical Model
  Space: Euclidean 3D
  Time: laboratory coordinate time
  Axiom: no physical signal exceeds c
  Axiom: detector response requires signal arrival

Frame: Narrative Knowledge Model
  Order: authored scene order
  Axiom: the narrator may reveal any fact immediately
  Axiom: character knowledge changes only through declared channels
```

4.3 Branches preserve alternatives

A fork occurs when a consequential commitment changes. The parent branch remains intact, while the child inherits its history and records the point of divergence. This supports comparisons across varied axioms without rewriting earlier work.

```
Shared scenario
  |-- Branch A: finite propagation; laboratory frame
  |-- Branch B: instantaneous bonded communication
  |-- Branch C: narrator knows instantly; characters do not
  |-- Branch D: detector event authored without mechanism
  `-- Branch E: unresolved - no rule determines receipt
```

The same capability applies at a deeper level. A future fork may change the meaning of identity, remove global state, replace events with constraints, or abandon the present generalism in favor of another descriptive architecture.

4.4 Inheritance, override, and exception

Operation	Meaning	Required record
Inherit	Use the parent rule unchanged.	source branch and version
Override	Replace a rule within a declared scope.	old rule, new rule, scope, reason
Exception	Suspend a rule for one event or relation.	event, affected rule, consequences
Fork	Create a separate continuation.	fork point and changed commitments
Merge attempt	Construct a child from two branches.	translation contracts and residue

5. Classes of Relation and Resolution

5.1 Spatiotemporal relations

These relations govern processes through location and duration. Examples include motion, collision, wave evolution, fluid transport, signal propagation, biological growth, mechanical stress, and embodied action. A strict physical frame may require local interactions, finite speeds, conservation laws, and numerical stability.

5.2 Temporal but not necessarily spatial relations

These relations govern order, delay, recurrence, development, memory, or procedural dependency without requiring a meaningful geometric path. A promise can remain in force until fulfilled. A musical theme can return after an earlier statement. A plan can require one stage before another. A remembered event can affect a later decision even when the memory is not modeled as a physical object traveling through scene space.

5.3 Spatial or structural but not necessarily temporal relations

These relations describe containment, topology, adjacency, layout, part-whole organization, maps, diagrams, type hierarchies, or geometric structure. They may remain static or may become the substrate for later transformations.

5.4 Atemporal and aspatial relations

Logical implication, mathematical correspondence, divisibility, category membership, formal equivalence, semantic opposition, and symbolic association need not receive a physical duration or location. The environment should not force them to impersonate signals moving through a world. They may still influence a world-model branch through an explicit compiler, solver, policy, or authoring bridge.

5.5 Interpretive and normative relations

Some relations assign significance, obligation, permission, salience, trust, identity, or value. They can guide an agent, author, or institution without behaving as physical forces. Their effects become world changes only when a rule connects interpretation or norm to action.

Relation family	Requires space?	Requires time?	Typical result
Spatiotemporal	usually	usually	movement, propagation, local interaction
Temporal	not necessarily	yes	ordering, delay, recurrence
Structural	not necessarily	not necessarily	arrangement, containment, topology
Formal	no	no	derivation, inclusion, contradiction
Interpretive	no fixed requirement	often	meaning, policy, significance
Cross-frame	depends on both frames	depends on mapping	translation or intervention

5.6 Resolution states

A description does not need to become one complete executable universe. The environment should record how far it resolves:

World-resolved: The active states, relations, and rules determine an executable continuation.

Partially resolved: Some processes execute while other claims remain authored, symbolic, or underspecified.

Multi-frame resolved: Several executable frames coexist and exchange information through declared bridges.

Branch-relative: Each branch resolves internally, but no lossless merge exists.

Interpretively open: Events resolve, while their meaning remains plural or disputed.

Formally unresolved: The description lacks enough rules to determine a continuation.

Contradictory by declaration: The user intentionally preserves incompatible commitments.

> **Paradox discipline**
> Plural frames, incomplete mappings, and unresolved descriptions are not automatically paradoxes. A paradox or contradiction requires incompatible commitments to apply to the same scoped claim under the same active rules.

6. Mixed-Mode Processes: Instantaneous and Finite-Speed

6.1 Why mixed modes matter

A sufficiently expressive storytelling and process tool must be able to place different forms of relation in the same authored sequence. A narrator may disclose a fact instantly to the audience. A mathematical constraint may apply without elapsed time. A character may experience a symbolic association. Meanwhile, a physical signal may require ten seconds to reach a detector. The process description should preserve all of these without pretending they share one propagation mechanism.

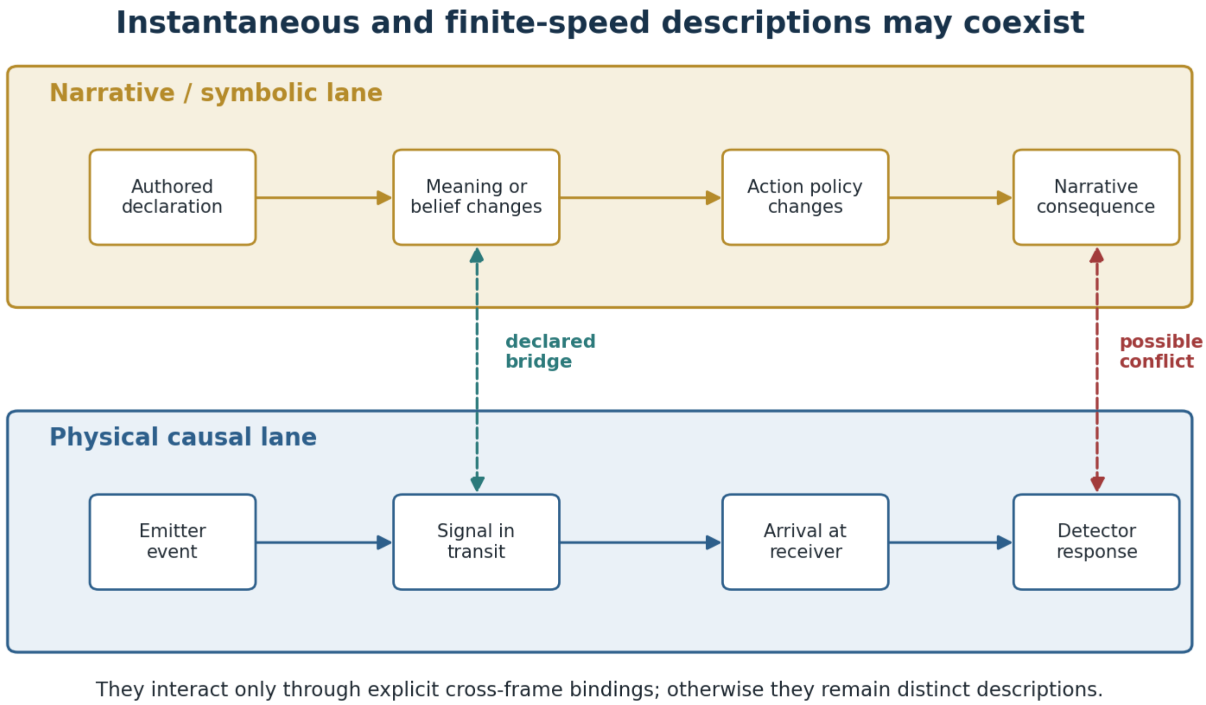

Figure 2. Mixed-mode descriptions remain in separate lanes until a declared bridge makes one relevant to the other.

6.2 Cross-frame bridges

A bridge states how something in one frame becomes input to another. The bridge may be causal, interpretive, observational, computational, editorial, or fictional. It should declare what it preserves and what it adds.

- **Observation bridge:** A physical detector event becomes evidence available to an observer or analysis frame.
- **Interpretation bridge:** Evidence changes an agent's belief, salience, or narrative understanding.
- **Action bridge:** A belief, plan, or norm triggers an embodied action in a world-model frame.
- **Authorial bridge:** The author directly changes a state or event, with the intervention marked as authored.
- **Translation bridge:** A variable or structure in one model maps to a counterpart in another.
- **Narrative override:** A story event is asserted despite lacking or violating the current world-model mechanism.

6.3 Worked mixed-mode example

Consider a source and a detector separated by ten light-seconds. At Turn 4, the source emits a signal. In the physical branch, the detector cannot register that signal until Turn 14. At the same moment, a narrator may state, "The detector has already been chosen as the place where the truth will appear." That statement can change narrative significance immediately without changing the detector's physical state.

Several continuations are now possible:

- The narrator knows the future, while the detector remains physically inactive until Turn 14.
- A character hears the narration through an explicit nonphysical channel and changes behavior, while the sensor state remains unchanged.
- An authorial override sets the detector to active at Turn 4; the system records a departure from the finite-propagation branch.
- A new branch introduces an instantaneous bonded channel and calculates its consequences.
- No bridge exists, so the narrative statement and physical process coexist without direct interaction.

A contradiction appears only if the same detector registration is simultaneously asserted to require physical arrival and to occur before arrival, with both rules active in the same scope and no declared override. The engine should then surface a rule conflict rather than erase either account.

6.4 Anomaly classes

Class	Meaning	Recommended treatment
Rule contradiction	Two active rules prohibit and require the same scoped result.	flag, fork, prioritize, narrow scope, or preserve intentionally
Frame mismatch	Claims use different clocks, spaces, or meanings.	translate before comparison
Underspecification	No rule determines the next state.	request authorship or leave open
Observation mismatch	A model prediction differs from a registered outcome.	record residual and revise or fork
Narrative override	An authored event bypasses the active mechanism.	preserve event and mark accountability boundary
Numerical artifact	A solver or interpolation creates the appearance of an event.	mark provenance and refine resolution
Intentional impossibility	The branch values contradiction or impossibility as content.	retain without presenting it as physically derived

7. Exemplar: The Causal-Resolution Layer

7.1 Why this exemplar belongs here

Ordinary timelines often treat each numbered frame as a universal update boundary. That convention works for many visual tasks, but it can quietly allow effects to occur everywhere at once. A more accountable process model needs a layer between the visible timeline and the event graph: a causal-resolution layer that determines whether influence has had enough modeled time to cross the relevant separation.

This exemplar demonstrates the larger generalism in a strict form. The objects are related through distance and channels. The relations receive propagation rules. Those rules constrain which events may follow. Other branches may replace the rules, but the consequences of each replacement remain inspectable.

7.2 Maximum reach per turn

For a channel with propagation speed v_ch and a turn duration Delta t, the maximum distance that an influence can traverse during that turn is:

$$d_{\mathrm{max}} = v_{\mathrm{ch}}\,\Delta t$$

When the selected channel represents electromagnetic propagation in vacuum, v_ch equals c. Sound, fluid waves, mechanical stress, material excitations, agent messages, or speculative channels may use different speeds or propagation laws.

7.3 A turn is a causal interval

A turn should represent a bounded interval rather than merely the next visual frame:

```
Turn n
|-- Start time: t_n
|-- Duration: Delta t_n
|-- Initial branch state
|-- Signals already arriving
|-- Permitted local interactions
|-- Events generated during the interval
|-- Signals launched during the interval
|-- Signals still in transit at the end
|-- Samples and observations recorded
`-- Resulting state at t_(n+1)
```

The turn number identifies the interval being examined. The turn duration determines how much modeled process can occur inside it. A short turn permits limited displacement and propagation. A long turn permits more development but may obscure event order unless the solver subdivides the interval.

7.4 Propagation edges have duration

Suppose a source and detector sit ten light-seconds apart and the turn duration is one second. An emission during Turn 4 remains in transit through successive turns. The detector interaction first becomes physically eligible at Turn 14.

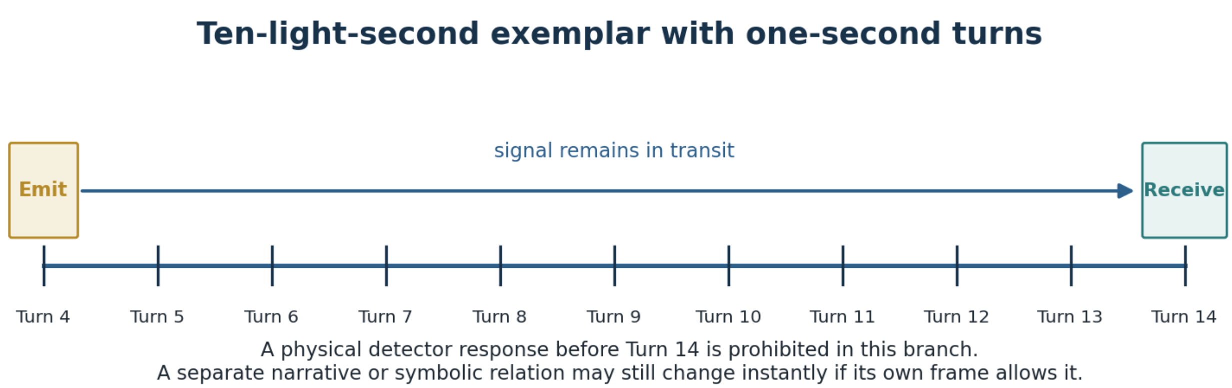

Figure 3. Finite propagation creates a durable in-transit state instead of an instantaneous event-graph edge.

For a constant-speed channel across distance d:

$$T_{\text{prop}} = \frac{d}{v_{\text{ch}}}$$

A receiving effect becomes eligible only after propagation and any modeled response latency:

$$t_{\text{effect}} \geq t_{\text{cause}} + T_{\text{prop}} + T_{\text{response}}$$

For a variable local propagation speed along a path gamma, the travel time may instead require integration:

$$T_{\text{prop}} = \int_{\gamma} \frac{ds}{v_{\text{ch}}(x, t, \sigma)}$$

7.5 Causal depth, not arbitrary event count

A turn may contain many independent local events. What it should limit is the depth of causal chains that traverse space or consume response time. The following sequence may be invalid inside one unsplit turn if its accumulated travel and response requirements exceed the interval:

```
A affects B
B immediately affects C
C immediately affects D
```

The engine can preserve order through microsteps:

```
Turn 12
|-- Substep 12.1: A influences B
|-- Substep 12.2: B begins responding
|-- Substep 12.3: B emits toward C
`-- Turn end: the signal to C remains in transit
```

This prevents the software update loop from creating superluminal or infinite cascades merely because all bindings were evaluated during one frame.

7.6 Particle-like descriptions

A particle branch may store a sampled worldline with identity, position, momentum, energy, internal state, received interactions, emitted interactions, and the next sample. For a massive particle in a relativistically constrained branch, displacement during a turn remains below c Delta t. A nonrelativistic branch may use much lower practical speeds while retaining c as an absolute causal ceiling.

Samples associated with one identity are not automatically separate objects. They represent recorded manifestations along one modeled history. The renderer should distinguish sampled positions, interpolated paths, predicted paths, and observed detector events.

7.7 Wave and field descriptions

A wave requires a distributed field state rather than a rigid object transform. Its branch may store amplitude, phase, frequency, wavelength, polarization, propagation direction, source history, medium, and boundary interactions. At each solver step, the field advances through the selected governing equation.

The interface should permit three simultaneous readings:

- Spatial view: amplitude, phase, or intensity distributed through space.
- Temporal view: the value recorded at one selected location over time.
- Causal view: source event, propagation, boundary response, and detector event.

7.8 Particle, wave, field, and measurement frames may coexist

The exemplar should not force the user to decide universally that a process is a particle or a wave. A branch may preserve several readings: localized emission and detection, distributed amplitude and phase, continuous local coupling, and discrete registered outcomes. Their coexistence does not prove one metaphysical identity. It permits comparison of what each frame exposes and what each leaves implicit.

7.9 Variable temporal grain

```
Large narrative segment
        -> physical turn
        -> solver step
        -> collision or interaction substep

Example:
Scene segment:       5 seconds
Simulation turn:     1/60 second
Wave solver step:    1/600 second
Collision substep:   adaptive
```

The visible timeline may remain simple while the causal engine evaluates finer intervals underneath it. A branch may also use proper time, logical transition index, local clocks, or event-order positions. These values should remain explicit rather than being flattened into one universal timeline.

7.10 Causal cones and reachable regions

Selecting an event should reveal the region it may influence under the active model. In three-dimensional space, the boundary may appear as an expanding sphere. In a spacetime view, it may appear as a cone. Anisotropic media may produce irregular fronts. Graph-based spaces may produce expanding neighborhoods rather than metric spheres.

The visualization should distinguish reachable, not-yet-reachable, unreachable, blocked, attenuated below detection, uncertain, and causally available but unactivated states.

7.11 Proposed causal-resolution panel

```
TIME MODEL

Traversal time                  00:04.200
Current turn                    252
Turn duration                   0.01667 s
Reference clock                 Laboratory frame
Selected channel                Electromagnetic
Propagation ceiling             c
Maximum reach this turn         4,996,540 m
Solver substeps                 10

[x] Enforce causal propagation
[x] Show causal cones
[x] Mark interpolated states
[x] Preserve signals in transit
[x] Display response latency
[x] Record branch provenance
```

For an air-pressure wave, the same panel might show a propagation speed of 343 m/s and a maximum reach of approximately 5.72 m during a 0.01667-second turn. The interface should state the assumptions and medium rather than treating the displayed number as universal.

7.12 Scope of the exemplar

The causal-resolution layer does not claim to solve full relativistic dynamics, quantum measurement, continuum field theory, or every numerical stability problem. It establishes a disciplined discrete architecture in which effects do not silently outrun their causes under the selected rule set. A later branch may add frame-dependent simultaneity, proper time, Lorentz transformations, curved-spacetime causal structure, or nonlocal models.

8. Comparison Across Varied Axioms

8.1 Comparison is not forced unification

Branches can be compared without being merged. Two descriptions may agree on an observation while disagreeing about intermediate states. They may preserve similar structures with different vocabularies. They may share no exact translation at all. The comparator should record the scope of agreement rather than converting resemblance into identity.

8.2 Comparison dimensions

Outcome: Do the same detector registrations, scene events, or formal conclusions occur?

Temporal: Do the branches agree on order, duration, delay, recurrence, or simultaneity?

Spatial: Do they agree on location, path, support, containment, or adjacency?

Causal: Do they assign the same ancestors, channels, and enabling conditions?

Structural: Do they preserve comparable invariants, topologies, feedback loops, or organization?

Identity: What counts as the same entity across transformation, sampling, or regeneration?

Interpretive: Which meanings, values, or narrative roles are assigned to the same event?

Accountability: Which steps are measured, derived, approximated, invented, or overridden?

User value: Which branch better serves the present artistic, scientific, explanatory, or practical goal?

8.3 Epistemic and authorial status

Status	Meaning
Authored	Directly supplied by a user, imported text, or declared scenario.
Axiomatic	Accepted as a rule or premise within the active frame.
Calculated	Produced by applying the declared model or solver.
Sampled	Recorded at a selected time, event, or resolution.
Interpolated	Constructed between samples; not automatically observed.
Observed	Registered by a detector, witness, agent, or imported measurement.
Inferred	Estimated from observations or other states.
Predicted	Projected beyond the current state.
Translated	Mapped from another frame with a declared contract.
Interpretive	Assigned meaning, significance, or label.
Fictionally asserted	Declared true within a story without an implemented mechanism.
Overridden	Produced by an explicit suspension or replacement of a rule.
Conflicted	Required and prohibited by different active commitments.

8.4 Translation contracts

An attempted integration should create an explicit mapping rather than silently combine branches. A translation contract should state the source and target variables, the domain of validity, information lost, reversibility, approximation error, calibration data, and unresolved residue. It may be exact, approximate, many-to-one, scale-limited, observation-equivalent, structurally analogous, interpretive only, or unresolved.

```
Source branch: Wave field
Target branch: Detector-event model
Mapping: field intensity above threshold -> detector registration
Preserved: registration time within tolerance
Lost: phase and distributed field state
Status: observation-equivalent for this detector configuration
Residue: no one-to-one particle path is supplied
```

8.5 Comparison report example

```
Branches agree on:
  - detector activation
  - arrival interval
  - total transferred energy within tolerance

Branches disagree on:
  - intermediate localization
  - state representation
  - whether a persistent particle identity exists

Narrative branch adds:
  - symbolic significance of the detector
  - foreknowledge available to the audience

Scientific accountability boundary:
  - physical derivation ends at the detector event
  - symbolic meaning enters through an interpretive bridge
```

9. Biological-Cybernetic Default Framing

9.1 The coupled authoring pair

The project is being produced and used through a coupled biological and computational process. A biological participant perceives, selects, imagines, values, and revises. A computational system stores, transforms, compares, renders, and executes descriptions. Each output becomes new input to the coupled loop.

```
Biological participant
  -> authors, observes, evaluates
Descriptive environment
  -> stores, executes, compares, renders
Result
  -> changes attention, understanding, and intention
Biological participant
```

This framing explains why the interface benefits from native categories such as ground, biological regulation, reach, attention, understanding, emotion, action, social relation, identity, moral orientation, cultural framing, and existential significance. These categories describe common aspects of the human-system loop. They do not require every simulated world to behave as an organism or every agent to implement one cognitive theory.

9.2 Suggested native relation vocabularies

Loop or zone	Guiding question	Possible native relations
Ground	Where am I?	support, gravity, orientation, direct contact, local frame
Biological	How am I?	interoception, hunger, fatigue, temperature, homeostasis
Reach	What can I physically do?	range, manipulability, effort, obstruction, affordance
Attention	What matters now?	salience, filtering, prioritization, inhibition, focus
Understanding	What is happening?	prediction, pattern, model, mismatch, revision
Emotion	What is its significance?	valence, arousal, urgency, attraction, avoidance
Action	What should happen next?	policy, plan, motor execution, outward change
Social	What do others communicate or expect?	role, recognition, trust, obligation, conflict
Identity	What continuity organizes the self-description?	memory, narrative, role, anticipated future
Moral	What is permitted, harmful, or required?	duty, value, fairness, responsibility, boundary
Cultural	Which inherited frameworks shape meaning?	language, symbol, ritual, convention, institution

Loop or zone	Guiding question	Possible native relations
Existential	What larger orientation gives meaning?	purpose, mortality, belonging, awe, finitude

9.3 Cross-disciplinary category packs

The system may acknowledge families drawn from cognitive science, neuroscience, biology, cybernetics, systems theory, psychology, philosophy, active inference, morphogenesis, autopoiesis, process thought, relational ontology, narrative identity, and enactive constraint closure. These should enter as optional relation packs, prompts, rule templates, or interpretive vocabularies. Shared vocabulary should not be treated as proof that the theories are identical.

10. Design Requirements and Native Primitives

10.1 Required primitives

Primitive	Role
Identity	A reference used to associate states, events, samples, and relations.
State	A declared configuration at a time, turn, event, or solver step.
Relation	A typed connection among identities, states, events, frames, or descriptions.
Rule	A scoped contract that governs a relation or relation class.
Transformation	A change from one state or organization to another.
Event	A bounded occurrence, threshold crossing, registration, or transition.
Binding	A relation whose conditions make a response eligible.
Propagation	A process by which influence travels before another response becomes possible.
Turn	A bounded interval used to enumerate and constrain updates.
Sample	A recorded representation at a selected resolution.
Observation	A registration produced by a detector, witness, agent, or imported record.
Frame	A declaration of vocabulary, axioms, clocks, spaces, and standards of consequence.
Branch	An inherited but isolated continuation of a frame and its history.
Bridge	A rule that permits information or intervention to cross frames.
Translation	A mapping between variables, events, structures, or observables.
Residue	A difference or loss that remains after comparison or translation.
Anomaly	A flagged mismatch, contradiction, override, or unresolved condition.
Provenance	The source and status of each authored, derived, observed, or translated claim.

10.2 Rule contracts

Rules should be inspectable objects rather than hidden code paths. A rule contract should expose at least:

```
Rule identity and version
Domain and frame
Participating roles
Required relations
Preconditions and trigger
Transformation or conclusion
Propagation method and latency
Resource cost or conserved quantity
Forbidden outcomes
Uncertainty or tolerance
Observation products
Exceptions and priority
Parent theory, source, or authored origin
Branch provenance
```

10.3 Event and relation inspector

Selecting any event or relation should answer:

- What frame and branch contain it?
- Which identities and states participate?
- Which rule made the transition eligible?
- What source, channel, delay, or bridge was used?
- Which facts were authored, calculated, observed, inferred, or interpreted?
- Which alternative branches produce different results?
- Where does the chain leave a selected scientific or mathematical basis?
- What residue or anomaly remains unresolved?

10.4 Multiple clocks and ordering systems

The system should support a presentation timeline alongside branch-specific clocks and ordering systems. A record may include scene time, laboratory time, proper time, solver step, event-order index, narrative order, logical dependency order, and uncertainty interval. A relation uses only the dimensions relevant to its frame.

10.5 Conflict handling

When rules conflict, the engine should not silently choose a winner. It should offer at least five operations: narrow a rule's scope, assign priority, add an explicit exception, fork the branch, or preserve the contradiction. The chosen operation becomes part of provenance.

10.6 Interface modules for this subdocument

Frame and Branch Stack: Shows inherited axioms, overrides, active clocks, spaces, and fork history.

Relation Inspector: Shows roles, rules, bindings, bridges, and current eligibility.

Process Timeline: Shows narrative segments, turns, solver steps, event order, and signals in transit.

Causal Resolution Panel: Shows channel speed, maximum reach, response latency, and causal cones.

Accountability Ledger: Shows authored, axiomatic, calculated, observed, translated, and overridden steps.

Comparison Matrix: Compares outcomes, timing, cause, structure, interpretation, and user value across branches.

Anomaly Panel: Surfaces contradictions, mismatches, under-specification, and intentional impossibility.

Bridge Editor: Defines how information or action crosses frames.

10.7 Minimal data shape

```
ProcessRecord
  id
  frame_id
  branch_id
  identities[]
  states[]
  relations[]
  rules[]
  events[]
  observations[]
  bridges[]
  translations[]
  anomalies[]
  provenance[]

Relation
  type
  participants[]
  dimensions_used[]
  active_rules[]
  scope
  status

Event
  authored_order
  branch_time
  solver_step
  causes[]
  effects[]
  eligibility
  epistemic_status
```

10.8 Audit rules for the causal exemplar

- No finite-speed effect occurs before permitted arrival.
- No massive particle displacement exceeds the active branch ceiling.
- No causal chain hides more propagation and response time than the containing interval permits.
- Signals in transit remain represented until arrival, cancellation, or dissipation.
- Interpolation remains visibly distinct from observation.
- A branch merge requires declared mappings and residue.
- A global solver update is not labeled as local physical propagation unless a rule justifies it.
- A narrative override remains visible rather than being rewritten as a calculated physical event.

11. Extension and Forkability

11.1 Future exemplars

The present document fully develops only the causal-resolution exemplar. The same generalism can later support additional subdocuments without expanding this one into a catalogue. Plausible future exemplars include:

- A biological or morphogenetic process with nested regulation, boundary maintenance, and regeneration.
- A cognitive-agent process using attention, understanding, emotion, action, and social relation loops.
- An atemporal mathematical or logical process in which entailment replaces propagation.
- A normative or legal process in which permissions and obligations govern possible actions.
- An impossible-geometry world with branch-specific spatial axioms.
- A narrative agency model in which symbolic, remembered, and anticipated events shape action.
- A multi-model scientific comparator with explicit translation contracts and empirical residuals.

11.2 Forking the current exemplar

The causal exemplar may itself fork into alternative physical and fictional branches: finite propagation in a preferred laboratory frame, relativistic observer-dependent timing, anisotropic media, graph-distance propagation, delayed narrative causation, instantaneous bonded channels, retrocausal rules, or deliberately inconsistent worlds. Each branch should preserve its own consequences and declare any bridge to the others.

11.3 Forking the generalism

A deeper fork may change the project's descriptive primitives. A future architecture could remove identity persistence, define processes entirely through relations, replace event sequences with constraint networks, or treat frames as dynamically generated. The current design should support export, translation, and coexistence rather than making its own categories impossible to revise.

EXTENSION PRINCIPLE

Add new exemplars as focused subdocuments. Fork the generalism when a different descriptive basis becomes useful. Preserve the source frame, translation, and residue rather than rewriting the project history.

Conclusion

This subdocument establishes a focused method for combining unconstrained description with accountable process modeling. The tool may accept a story, an absurd premise, a mathematical relation, an interpretive claim, a physical model, or a mixture of all of them. Those descriptions remain free at the point of authorship. They acquire local necessity when rules bind their relations inside declared frames.

That structure permits instantaneous, atemporal, and aspatial relations to coexist with processes constrained by distance, duration, and finite propagation speed. They do not need to collapse into one world model, and their coexistence does not by itself produce a paradox. Explicit bridges determine when they interact. Conflicts appear only when incompatible commitments govern the same scoped claim.

The causal-resolution exemplar demonstrates the method at a demanding level. It preserves signals in transit, limits causal reach, distinguishes samples from interpolation and observation, supports particle and wave readings, and allows branch-relative alternatives. It also remains only one exemplar. Other subdocuments may develop biological, cognitive, logical, normative, or deliberately impossible processes through the same or a forked generalism.

Most importantly, the environment can compare descriptions across varied axioms without deciding that one kind of value erases another. A fictional branch may matter more to its writer than a physically realizable one. A scientific user may instead require an exact account of where derivation ends and added premise begins. The system serves both by preserving provenance, scope, consequence, and residue.

FINAL DESIGN STATEMENT

A description may be freely imagined. A relation may be rigorously governed. A frame declares where the rule applies. A branch preserves alternatives. A bridge permits interaction. A comparison reveals agreement and residue without forcing one account to become the measure of every other.

END OF SUBDOCUMENT